

An Intelligent Security Framework for Industrial IoT Using Swarm Based Optimized Ensemble Machine Learning Model

K. Eswara Rao¹, Ch. Ravikishore², Ajay Prasad Kushwaha³, V. Jyotsna Devi⁴, Dr. Trinadha Burle⁵ and Pondreti Ramkishor⁶

^{1,2,3,6}Dept. of CSE, Aditya Institute of Technology and Management, Tekkali, Srikakulam, INDIA

⁵Dept. of EEE, Vignana's Institute of information technology (A), Duvvada, Visakhapatnam, INDIA

⁴Dept. of IT, Aditya Institute of Technology and Management, Tekkali, Srikakulam, INDIA

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Abstract: The worldwide expansion of the Internet of Things has been characterized by significant growth, as seen by the widespread use of diverse devices in domains such as housing environments, transportation systems, healthcare facilities, and industrial sectors. The introduction and integration of the IoT concept in industrial environments has resulted in significant modifications to the architecture of Industrial Automation and Control Systems (IACS) and the widespread interconnectivity of various industrial systems. The outcome of this development is often known as the Industrial IoT (IIoT), which eliminates the obstacle of linking IACS with separate traditional Information and Communication Technology (ICT) platforms. In contemporary times, the IoT has begun to affect our individual lives and extend its influence beyond our immediate surroundings, establishing a foundation for imminent cyber-attacks targeting the IoT. The extensive utilization of the IoT has produced a productive ground for potential assaults against IoT systems. Machine learning (ML) algorithms have enhanced wireless communication security in IIoT-based systems and addressed various cyber security issues. Hence, this study presents an improved ensemble model for an intrusion detection system that utilizes Particle Swarm Optimization (PSO) to classify hostile behaviors in network data within IIoT. The performance of the proposed design was evaluated using precision, recall, F1 score, accuracy, F2 score, and ROC-AUC metrics on the X-IIoTID Dataset, an IIoT-based cyber security dataset. The obtained findings were compared to those of other recent state-of-the-art ML models, and it was observed that our model exhibited superior performance.

Keywords: Machine Learning, IoT, IIoT, Attack, Security, PSO, Optimization, Intrusion Detection System (IDS)

1. INTRODUCTION

The Internet of Things (IoT) is a system of interconnected contraptions that support the interaction between "things" and allow the development of distributed applications using more complex structures like computing. The IoT is constantly advancing, with a growing number of connected devices. As a result of this expansion, we accumulate more data, evaluate it, and use it to develop intricate algorithms. These algorithms enable the automation of various operations, resulting in increased efficiency. Under the IoT superclass, the IIoT subclass is one of the categories that refer to IoT technologies utilized in industrial contexts, specifically in manufacturing plants. The IIoT refers to using IoT technology specifically in the industrial sector, particularly for advanced manufacturing purposes [1]. The IIoT was a network of sensors and self-contained devices communicating with industrial applications via the internet/network. This network allows for data collection, analysis, and production optimization, increasing

efficiency and lowering manufacturing and service costs. IIoT technology could assist industrial companies in increasing productivity by improving predictive maintenance plans, i.e., the state and functionality of equipment, to predict when it may occur. The industrial internet reference architecture may be a reference for developing sophisticated systems in the IIoT domain [2]. According to statistics from Oxford Economics, the IIoT can potentially affect sectors that contribute to 62% of the Gross Domestic Product (GDP) in the G20 nations. Therefore, implementing IIoT in various industries is expected to be the primary catalyst for productivity and innovation in the next decade [3]. Many industries can benefit from the IIoT, including those dealing with autonomous vehicles and healthcare, optimizing the efficiency of machines, minimizing human mistakes, improving distribution and logistics, and reducing the overall number of accidents. The emergence of IIoT applications has led to a proportional surge in security issues and cyberattacks [4]. Prominent threats in IIoT systems include

man-in-the-middle attacks, denial-of-service attacks, device exploitation, data interception and manipulation, firmware flaws, and software vulnerabilities. The goal is to make IIoT systems safer, so we should look at some traditional things: regularly updating and patching firmware and software, encrypting network communications, putting in intrusion detection and prevention systems, teaching employees best practices for cybersecurity, making plans for how to handle incidents and recover from them, and doing regular security checks and penetration tests. Over the last decade, several cyberattacks have targeted the IIoT, impacting software and hardware components. These assaults have impacted various system elements, including the status of pumps and sensors [5] [6]. To safeguard the security of IIoT nodes and communication networks, many enterprises employ Intrusion Detection Systems (IDS) to defend against vicious attacks [7]. Hybrid IDSs aim to improve upon the shortcomings of signature and anomaly-based IDSs. Traditional intrusion detection systems are prone to producing misleading results due to their high false-positive rates and poor detection accuracy. Machine learning (ML) has greatly facilitated the creation of advanced solutions that streamline processes, enhance decision-making, and optimize operations by leveraging the capabilities of artificial intelligence (AI) [8]. ML is crucial for contextual analysis in the IIoT since it improves the system's ability to analyze and resolve network difficulties and problems with IoT devices over time [9]. ML approaches are the most often used techniques for conducting intrusion detection [5]. ML may enhance experimental learning and decision-making in different systems by boosting their abilities and capacity without explicit programming [10]. Contemporary analyses have shown that ML approaches can effectively close some security holes and make anomaly-based detection models work better [11] [12] [13]. Implementing the ML model in the IIoT network encounters several obstacles, including limited processing capacity and data analytics techniques [14]. ML algorithms classify anomaly events like assaults and IIoT hardware failures. Linear Discriminant Analysis (LDA), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Logistic Regression (LR), Naïve Bayes (NB), and CART are some of the tools that are used to divide malicious activity into different groups. Conclusion: CART and NB exhibit superior accuracy, precision, recall, and F1 score [15].

A. Background Study

Traditional ML approaches, such as Bayesian Belief Networks (BBN) and SVM, have been employed in cybersecurity. IIoT generates immense amounts of data, necessitating the design of an ML-based system to meet its specific requirements. Modifying and scaling the ML model while ensuring minimal performance changes are possible. This necessitates using more advanced ML methods and optimization strategies to efficiently process vast amounts of data in the manufacturing industry. Industrial IoT combines operational technology (OT) and information technology (IT) to address safety and security concerns [16]. Han-

dling the IIoT dataset using ML could be a considerable challenge. We must train the model to assure the security and privacy of the information, which is a complex task. IIoT datasets are high-dimensional data, so it is challenging for ML algorithms to handle complex datasets and find the learning patterns for accurate predictions. Due to the diverse nature of the IIoT dataset, it is challenging for the ML model to handle. Sample collection of IIoT data is not always the same because it depends on network behavior. If there are too many or too few data points, the model might be biased and unable to keep the IIoT dataset of majority and minority classes in balance. Normally, ML models are running in one setup parameter, which is by default. Often, a single parameter setup does not yield optimal performance due to the diverse nature of data in the IIoT. So, to avoid these limitations, hyperparameters of models are essential factors in ML. The user defines hyperparameters in ML to control the learning process and improve the model's learning rate. Iterative hyper-parameter tuning can also be tried with different parameters and values, changing how complicated the model is and how well it finds the IIoT network.

B. Research Contribution

ML model hyperparameter optimization improves performance by identifying the optimal combination of hyperparameter values within specific time frames. The automatic hyperparameter selection approach for determining the optimal network configuration is the most critical aspect of IIoT. Different optimization methods are used to find the best hyperparameters, such as random search, particle swarm optimization (PSO) [17], simulated annealing, the ensemble method, and grid search. Among all optimizations, PSO is the most popular. Studies have demonstrated that IIoT networks use PSO to fine-tune the hyperparameters for ML models, which significantly enhances their problem-finding capabilities. IIoT can utilize PSO to optimize communication protocols, schedule tasks in industrial processes, or enhance the overall efficiency and reliability of IIoT systems. To improve cyber-attack detection models, we can train them with route data sets and use ML and realistic PSO swarm optimization [18]. The hybrid PSO technique is accustomed to finding the optimal weights of hyperparameters to achieve greater accuracy in detecting attacks [19]. This research mainly offers the following main contributions:

- The IIoT ecosystem comprises diverse devices and sensors that generate substantial data. The ML technique is an important analytical tool for quickly analyzing large amounts of data and producing quick and effective real-time conclusions.
- An IDS employing PSO_XGBoost has been proposed to detect and minimize cyberattacks in an IIoT environment.
- To see how well-advanced ML ensemble methods and classic ML models work, this study looked at many

performance measures, such as recall, precision, F2 score, ROC-AUC, F1 score, and accuracy. The suggested model demonstrated a remarkable accuracy of 99.80%, outperforming existing advanced ensemble models and conventional ML methods.

The study effectively structures the subsequent portions. Section 2 provides a thorough analysis of the pertinent research. Section 3 presents an accurate account of the intended task. Section 4 provides a detailed explanation of the experimental procedure and summarizes the dataset. In addition, Section 5 provides a descriptive analysis of the simulation results. Section 6 provides a comprehensive summary of the research.

2. LITERATURE STUDIES

To effectively respond to harmful attacks, the IIoT must continually develop its security protection technologies to ensure its security. IIoT security poses a significant problem for industrial actors and academic research; integrating security measures like intrusion detection improves IIoT environment security. Relevant research keeps expanding and improving defenses against dangerous attacks, supporting industrial internet security's growth. This section delves into previous research on the IIoT IDS framework. Qaddoori et al. (2023) [20] developed a lightweight IDS for edge apparatus to detect MQTT-based attacks using ML models and the MQTT dataset. Multiple security mechanisms ensure reliable and secure data sharing between fog nodes and edge devices during updates, and the security assessment of the suggested security model is up to 99.69% accuracy. Agarwal et al. (2023) [21], the SDT (Splinted Decision Tree) method classifies the obtained features into different incursion groups, and this approach splits the feature space, allowing faster decision-making while maintaining accuracy. The suggested hybrid method worked better than other intrusion detection algorithms when the NSL-KDD dataset was used to train and estimate the model. Finally, the model is resilient to network traffic variations and accurately detects known and unknown intrusions. The detection accuracy of 98.9% was achieved when the IIoT incursions were recognized using the HDCA-SDT method in the NSL-KDD dataset. Momand et al. (2024) [22] suggested an attentionbased convolutional neural network (ABCNN) for intrusion detection in IoT systems. The attention mechanism of ABCNN facilitates learning in low-instance classes. The ABCNN framework's CNN optimizes critical parameters and detects harmful activities. The ABCN model is enhanced using the mutual information technique in the pre-processing phase to exclude the least significant dataset pieces. The datasets EdgeIoTset, IoTID20, ToN_IoT, and CICIDS2017 were used to assess ABCNN. The proposed model achieved an average accuracy of 99.81% across all datasets. It surpassed previous ML and deep learning (DL) models, with an accuracy of 98.02%, a recall of 98.18%, and an F1-score of 98.08%. Kaliappan et al. (2024) [23] developed an AI-enhanced trust framework for IoT intrusion detection to safeguard settings. The proposed IDS

leverages advanced AI to detect device activity anomalies and respond to emerging threats. A comprehensive IDS that identifies irregularities in real time is created using an Isolation Forest (IF) algorithm and Autoencoders (AE). IoT device behavior is modeled using LSTM and CNNs. The Reinforcement Learning-based Proximal Policy Optimization (PPO) dynamically reacts to anomalies, and the Bayesian Belief Network adapts to trust. The framework is compared against Conv-LSTM, cuLSTMGRU, ELET-IDS, Fed-Inforce-Fusion, and CNN-TSODE intrusion detection algorithms using IoTID20 and N-BaIoT dataset. The efficient framework has 98.25% detection accuracy, 96.8% recall, and 97.45% precision, outperforming baseline techniques. The AI-Enhanced Trust Framework is promising for detecting intrusions and creating secure IoT settings. New intrusion detection approach by Habeeb et al. (2024) [24]. This approach selects coarse and delicate features. To find feature set relationships, coarse selection uses correlation analysis. Step two employs Whale Optimization Algorithm (WOA) with Genetic Algorithm hybridization for exact selection. Fitness is determined using K-Nearest Neighbors (KNN) evaluation. The authors proposed the integration of WOA with a hybrid genetic algorithm to expand the search space and mitigate local optima challenges via crossover and mutation techniques. These traits help detect invasions. They deploy an ML classifier to detect network attacks and evaluate each classifier. They tested the classifier on the BoT-IoT 2020 dataset using 32 chosen characteristics to decrease computational complexity. This enhances system efficiency while balancing computational effectiveness and model performance. The experimental results demonstrate improved model accuracy compared to WOA and a reduced False Alarm Rate. Their CFWOA method showed 98.9% accuracy, while the GA-updated version had 99.5%. Rashid et al. (2023) [25] performed comprehensive tests on a novel Edge-IIoTset dataset. The performance assessment illustrates the dependability and efficacy of the suggested intrusion detection model, attaining an accuracy of 92.49%, comparable to the 93.92% accuracy of traditional centralized machine learning models using the federated learning approach. Tareq et al. (2024) [26] implemented a parallel approach using ML techniques, including LDA, SVM, SVM+LDA, and QDA, on the WUSTL-IIOT database and compared it with conventional approaches. The data is segmented into separate training datasets and trained concurrently. Experiments demonstrate that this parallel training system identifies and predicts cyber threats with greater accuracy. The detection speed of the parallel ML models was elevated, achieving an optimal accuracy of 100% with the SVM+LDA model. Eid et al. (2024) [27] examined the efficacy of six notable ML models, such as RF, DT, K-nearest neighbor, LR, SVM, and NB for IDS in IIoT settings. The assessment includes the impacts of data preparation methods, such as data normalization, feature engineering, recording, and handling missing data. The study investigates dataset balance, analyzing the impact of six distinct strategies on model performance. The research utilizes the specialized WUSTL-IIOT-2021 dataset, which

encapsulates the distinctive features of IIoT data. This study investigates the detection of multi-class attacks through an innovative SMOTE-based approach for balancing multi-class datasets, aimed at resolving issues of imbalance within the data. The results indicate that meticulous data preparation and astute dataset balancing reliably enhance the classification efficacy of the selected models in both binary and multi-classification endeavors. The RF method distinguishes itself by providing continuously superior performance with computational economy. Supervised ML was used to automate IIoT role engineering and fine-grained access control by Usman et al. (2023) [28]. Smart IoT systems connected to SCADA use an Extreme Learning Machine (ELM) and a carefully adjusted multilayer feed-forward Artificial Neural Network (ANN) to monitor user roles, privacy, and resource access. For the SCADA dataset, ELM was 89% accurate and ANN 96%. In this article, Naik et al. (2022) [29] compare the ML classification methods RF, LR, Light GBM, DT, and KNN to find things that do not fit the norm. PyCaret examined four IIoT datasets. Based on the on-demand versus response and e-filtration painting maintenance datasets, RF got 99.1% and 98.1% of the answers right, KNN got 94.4% of the answers right in semiconductor production, and RF got 93.98% of the answers right in HRSS. RF introduced the most excellent anomaly detection accuracy (98.15%). The RF method surpasses multivariate IIoT datasets. Table I presents the current research on the IIOT, which aligns with the current research goal.

A. Motivation to Proposed Model from Prior Studies

The reviewed literature presents significant advancements in IIoT intrusion detection, demonstrating a variety of approaches, including traditional ML, DL, hybrid optimization techniques, and AI-enhanced frameworks. Several studies have achieved high accuracy and efficiency in detecting intrusions. However, limitations remain, particularly in computational complexity, feature selection methodologies, adaptability to evolving threats, and real-time deployment feasibility. One of the key strengths of prior research is the integration of Machine Learning and Deep Learning techniques to enhance intrusion detection system accuracy. For instance, Qaddoori et al. (2023) [20] effectively utilized lightweight ML-based IDS for MQTT-based threat detection, achieving a high accuracy of 99.69%. Similarly, Momand et al. (2024) [22] demonstrated the effectiveness of attention-based CNNs, improving detection performance by leveraging mutual information for feature selection. Another notable approach is the hybridization of decision tree-based models, as proposed by Agarwal et al. (2023) [21], which enhances feature classification efficiency. Moreover, Habeeb et al. (2024) [24] introduced an advanced feature selection approach using WOA and GA, reducing computational complexity and optimizing model performance. Despite these advancements, certain limitations persist. Many studies rely on computationally expensive DL models, which may not be feasible for real-time IIoT environments with resource constraints. For

example, while Momand et al. (2024) [22] achieved 99.81% accuracy, the ABCNN framework's computational cost may hinder deployment in real-time industrial settings. Similarly, Habeeb et al. (2024) [24] improved feature selection but did not explicitly address the impact of hybrid optimization on processing speed and energy efficiency. Additionally, dataset dependency is another concern, as many studies, such as those by Kaliappan et al. (2024) [23] and Rashid et al. (2023) [25], rely on specific datasets, potentially limiting their models' generalizability across different IIoT networks. The proposed approach builds upon these existing methodologies by addressing their limitations and incorporating novel optimizations. Unlike prior works focusing on accuracy, our model optimizes accuracy and computational efficiency using an adaptive intrusion detection framework. It enhances feature selection to reduce model complexity while ensuring robust performance across diverse IIoT datasets. Moreover, by leveraging a novel optimization technique, our approach mitigates the challenges of local optima in traditional ML models, leading to improved adaptability against evolving cyber threats. Integrating an intelligent feature selection strategy further ensures an optimal trade-off between detection accuracy and computational overhead, establishing it as a practical option for real-world IIoT security implementations.

3. MATERIALS AND METHODOLOGIES

The optimization of hyperparameters has consistently been an essential component of prediction models. Here, we present the current body of research on these methodologies. Classification determines the category to which a new observation belongs, a data set containing prior observations and their corresponding categories. This research work aims to determine the significant factors that influence the outcome.

A. XG Boost

The technique, XGBoost [30], is highly effective for regression and classification problems. It serves as a compilation of effective schedules obtained from Kaggle competitions. XGBoost is an ensemble learning approach that uses the gradient boosting framework to build decision trees to make learners more efficient and improve their performance by considering residual values. When estimating the loss function, XGBoost uses a Taylor expansion, which differs from gradient boosting. The model's bias and variance are more balanced due to this method. More accuracy with fewer decision trees is a typical result when using XGBoost. Next, we will go over XGBoost. Let's consider a sample set with 'n' samples and 'm' features, denoted as

$$D = \{(x_i, y_i)\} \quad (|D| = n, x_i \in \mathbb{R}^m, y_i \in \mathbb{R})$$

where 'x' represents the eigenvalue and 'y' represents the actual value. The method aggregates the outcomes of 'N' trees to obtain the ultimate anticipated value, as shown in Eq. (1).

$$\hat{y}_i = \sum_{n=1}^N f_n(x_i), \quad f_n \in F \quad (1)$$

TABLE I. Summary of Existing Studies

Methods Used	Proposed Method	Performance Measures	Dataset	Benefit of the Study	Limitation of the Study	Year
DT, RF, GB	DT	Accu=99.68%, Prec=99.67%, Recall=99.68%, F1=99.66%	MQTTset dataset	It is effective against a wide range of internal and external attacks and strikes a balance between strong performance and high security	The security issue might arise in case of multi-tier IIoT structure.	2023
DL-IDPS, PRI-IDS, CNN-LSTM	HDCA-SDT	Accu=98.90%, Prec=98.70%, Recall=98.20%, FM=98.50%	NSL-KDD dataset	The HDCA-SDT framework efficiently detects and classifies abnormalities in complicated datasets by introducing feature extraction capabilities.	The IIoT new dataset, which could be acquired from the actual world, is quite challenging for this model.	2023
ANN, ELM	ANN	Accu=96.00%, prec=93.00%, F1=91.00%	SCADA Dataset	The highest level of flexibility may be quickly and efficiently achieved with fine-grained access control.	Not applicable for implementing attribute-based access control for IIoT use cases.	2023
Traditional CNN, Inception CNN	CNN	Accu=75.03%, Prec=73.67%, F1=73.66%	mobile network data	To develop security and anti-intrusion technologies for IIoT.	The model has not been validated in a complex multi-interaction network or applied in large-scale industrial processes.	2022
RF, LR, Light GBM, DT, KNN	RF	Accu=98.15%, AUC=99.48%, Recall=94.22%, Prec=97.92%, Kappa=94.82%, MCC=94.85%	HRSS dataset	Provides remarkable findings for multidimensional IIoT information.	In IIoT, complex and diverse data from various sensors can be combined to create an ensemble model for more accurate findings on large datasets.	2022
ANN, DT, SVM, RF, NB	ANN	Accu=99.97%, Prec=100.00%, Sensitivity=100.00%	TON_IoT dataset	Designing, structuring and implementing data structure and smart contracts in manufacturing.	Incorporating future technologies like 5G/6G communication systems in the network/protocol layer is a challenging task.	2022
SVM, KNN, DNN, CP-LNN	CP-LNN	Accu= 98.50%, Speci=98.20%, Prec=98.35%, FM=97.49%	Synthetic Dataset	CP-LNN was used to find the best way for keeping an eye on the industry and stopping accidents with simple control strategies	It will not be applicable for the diverse network.	2021
Neutro-(SVM) classification, LSTM	Neutrosophic (SVM)	Speci=99.90%, Prec=98.30%, F1=99.10%	Synthetic Dataset	To improve IIoT device experiences, extract features and provide a trust score for better decision-making.	Unable to address scalability issues in the industrial trust computation	2021
LSTM classification model	LSTM	Accu= 86.58%, Recall=64.84%, Prec=95.81%, F1=77.33%	ICS datasets	This technique minimises training time and enhances detection efficiency in diverse network contexts.	The problem may come from cross-correlating security warnings across different levels of ICS settings.	2021
SVM, DT, SVM, ANN, NB	RF	Accuracy=99.99%, MCC=96.81%, Sensitivity=97.44%, UR=52.56%	Synthetic Dataset	A feature priority ranking was conducted to identify the key elements that distinguish attack traffic from normal traffic.	-	2019

Where F is the set of decision trees or stumps

B. Particle Swarm Optimization (PSO)

The PSO [31] technique resembles a bird's behavior among a group of birds by creating a particle that lacks mass. This particle is characterized by its speed and location. Speed refers to the velocity at which an object travels, whereas position refers to its specific location in space. Each particle in the swarm independently searches for the best solution within its search space. It keeps track of this solution as its current individual extreme value. Additionally, each particle evaluates the extreme values of all particles in the swarm to determine the current global optimal solution. Based on this data, every particle modifies its velocity and location to maximize the efficiency of the swarm. This is the fundamental explanation of the conventional Particle Swarm Inference (PSO) algorithm. Consider a scenario with a population of 'M' particles in a search space with 'd' dimensions. Assume that at a certain time, 'T,' information is available on the particles' population. The position X_i is represented by a vector $\mathbf{X}_i = [x_1, x_2, \dots, x_d]$. The speed of V_i is represented as $\mathbf{V}_i = [v_1, v_2, \dots, v_d]$. The user's personal best position is represented as a vector $[pt_{i1}, pt_{i2}, \dots, pt_{id}]$. The global ideal position $\mathbf{p}_g$ is represented as $[\mathbf{p}_{1g}, \mathbf{p}_{2g}, \dots, \mathbf{p}_{dg}]$. Next, the velocity and location data of the particles is updated at time $T + 1$ using the following Eq. (2) and (3):

$$v_i^{(t+1)} = w \cdot v_i^t + c_1 \cdot r_1 \cdot (p_i^t - x_i^t) + c_2 \cdot r_2 \cdot (p_g^t - x_i^t) \quad (2)$$

$$x_i^{(t+1)} = x_i^t + v_i^{(t+1)} \quad (3)$$

The inertia weight 'w' is crucial in maintaining global and local exploration equilibrium. The learning parameters 'c1' and 'c2' control the length of each step to explore the populations and the global optimal positions. A function that follows a uniform distribution produces the random integers 'r1' and 'r2'.

4. PROPOSED WORK

This section aims to analyze the effectiveness of each optimization technique by comparing them to the X-IIoTID dataset. We employed an IIoT dataset, partitioned into a percentage of 20:80 for testing and training purposes. This research used the conventional ML approaches in conjunction with an ensemble method. After hyper-tuning it using four optimization strategies, such as the Genetic Algorithm (GA) and PSO, we looked at how much it costs to run. We compared different optimization techniques based on how hard they were to compute to find the best ways to change the XGBoost algorithm's hyperparameters. Figure 1 shows the sequential procedure of the research study. We choose the PSO parameters through empirical tuning, previous research [30] [31], and experimental validation to balance exploration and exploitation well. The swarm size is set to 30, ensuring a diverse search space. At the same time, the cognitive 'c1' and 'c2' coefficients are chosen as 1.5 and 1.7, respectively, to maintain a balance between individual learning and social influence. The inertia weight

'w' has dynamically adjusted between 0.4 and 0.9 to control convergence speed and prevent premature stagnation. The number of iterations, 100, is determined based on convergence analysis, ensuring computational efficiency while maximizing accuracy. To further validate our parameter selection, we have compared different PSO configurations and assessed their impact on XGBoost performance. The chosen parameters effectively optimized the hyperparameter space, outperforming traditional tuning techniques such as GA. Our method shows how flexible PSO can be when hyper-parameter tuning for ML models in IoT-based intrusion detection, ensuring high accuracy with little extra work on the computer. Table II shows the hyperparameter details of the meaning of XGBoost within the standard range. Algorithm 1 presents the algorithm of the proposed framework.

Algorithm 1 XGBoost with PSO

Input: Population M of m hyperparameter sets Ψ_i , Dataset $D = \{x_i, y_i\}_{i=1}^n$
Output: Optimal hyperparameter set $\mathbf{p}_g$
 Define PSO parameters: $c_1, c_2, r_1, r_2, v_i, w_{ps0}, max_gen, \mathbf{p}_t, \mathbf{p}_g$
 Set population size M and maximum generations max_gen
 Set values for $c_1, c_2, r_1, r_2, v_i, w_{ps0}$
for each particle $i = 1$ to m **do**
 Evaluate fitness: $f_i = f_{XGBoost}(\Psi_i, D)$
 Set personal best position $\mathbf{p}_t^{(i)} \leftarrow \Psi_i$
end for
 Set global best $\mathbf{p}_g$ as the best among all $\mathbf{p}_t^{(i)}$
for $k = 1$ to max_gen **do**
 for each particle $i = 1$ to m **do**
 Update velocity v_i using Eq. (2)
 Update position Ψ_i using Eq. (3)
 Evaluate fitness $f_{XGBoost}(\Psi_i, D)$
 if $f_{XGBoost}(\Psi_i, D) > f_{XGBoost}(\mathbf{p}_t^{(i)}, D)$ **then**
 $\mathbf{p}_t^{(i)} \leftarrow \Psi_i$
 end if
 if $f_{XGBoost}(\mathbf{p}_t^{(i)}, D) > f_{XGBoost}(\mathbf{p}_g, D)$ **then**
 $\mathbf{p}_g \leftarrow \mathbf{p}_t^{(i)}$
 end if
 end for
end for
return $\mathbf{p}_g$ as the optimal result

A. DATASET OVERVIEW AND SIMULATION ENVIRONMENT

There is much important information in the next section, including a complete summary of the data set, the simulation environment, and a comparison of the results from the proposed methods to those from previous research.

B. Simulation Setup

An HP Pavilion system with the following configurations has been employed for implementing the suggested

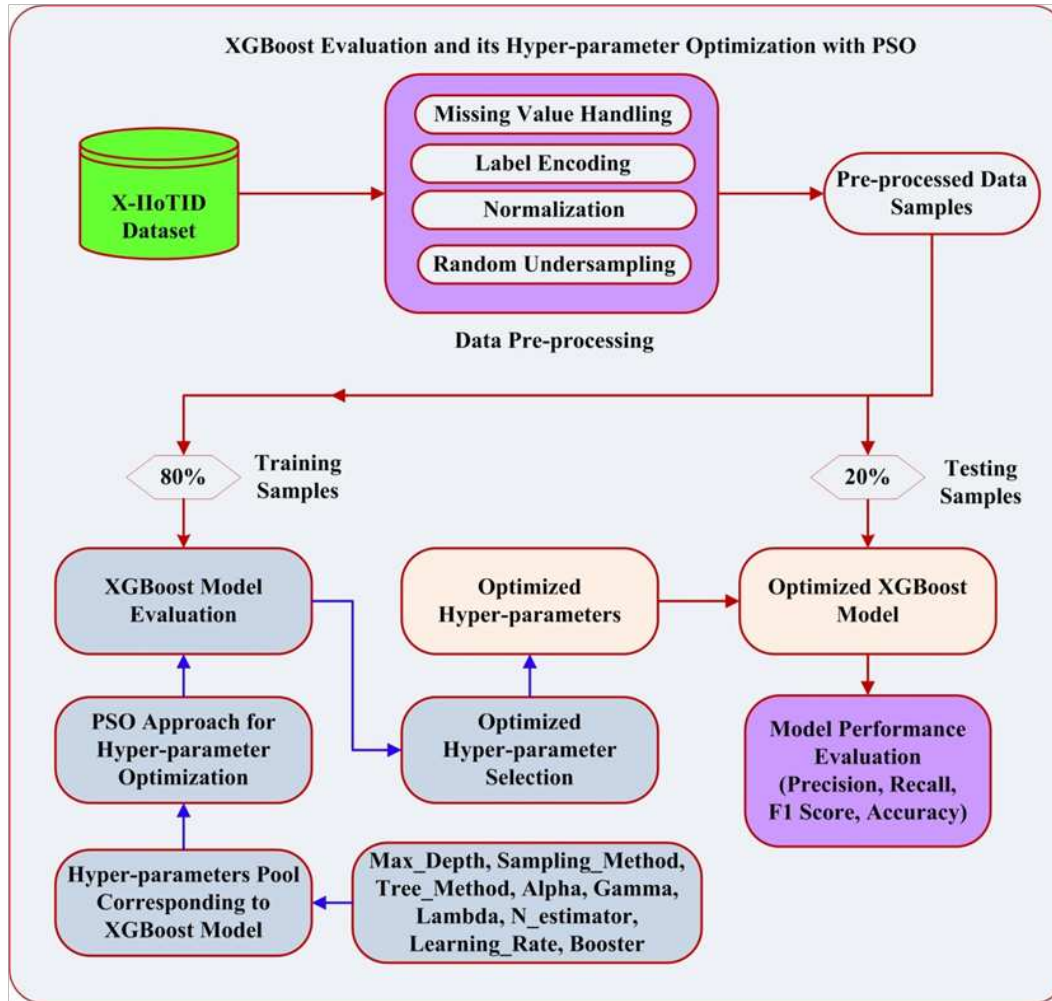


Figure 1. Proposed PSO_XGBoost Architecture

approach, in addition to many ML methods and ensemble learning techniques for comparison: The hardware prerequisites for the experiment consist of a 12th Generation Intel(R) Core(TM) i⁵ -1240P 1.70 GHz processor, 32 GB of RAM, Windows 11 Home, and an Intel Iris Xe graphics processing unit. The project employs the Python programming language, machine learning methodologies, and Google Colab Notebook for conducting experiments. The research also uses the open-source ML framework Scikit-learn, data analysis tools NumPy and pandas, and visualization tools Seaborn and Matplotlib.

C. Dataset Overview

The X-IIoTID dataset was collected from the IEEE-Data Port repository [32] for intrusion detection. Sixty-eight characteristics and 820,834 instances are included in the recent version of the X-IIoTID. The dataset comprises 421,417 recordings for normal instances and 399,417 for attack instances.

5. RESULT ANALYSIS AND DISCUSSION

To evaluate the effectiveness of the proposed research approach, the experimental procedure integrates many ML and ensemble learning methodologies. This section provides comprehensive information on the analysis of the findings. This research investigated various ML approaches, including LR, DT, SVM, and KNN, and ensemble methods, such as RF, AdaBoost, XGBoost, GA_XGBoost, and PSO_XGBoost. Also, recall, precision, accuracy, ROC-AUC score, F1 score, and F2 scores were used to rate the performance of all the studied approaches, including the suggested one. The PSO_XGBoost ensemble classification model stands out over all other models, obtaining an exceptional accuracy of 99.80%, F1 score of 0.9982, F2 score of 0.9980, precision of 0.9984, recall of 0.9981, and ROC-AUC score of 0.9994. The results demonstrate that the proposed method outperforms normal ML and group methods in terms of accuracy, precision, recall, F1 score, F2 score, ROC-AUC score, and other metrics. A suggested method in Table III shows all the evaluation tools used in

TABLE II. XGBoost Hyper-parameters Details

Hyper-parameter	Meaning	Standard Range
n_estimators	The number of boosting rounds (trees) in the ensemble	[50,100]
learning_rate	Shrinks feature weights at each step to prevent overfitting (shrinkage step).	[0.1, 0.3]
max_depth	Maximum depth of a tree; controls model complexity.	[3,12]
min_child_weight	Min. sum of instance weight needed in a child node; prevents overfitting.	[1,10]
lambda	L2 penalty on leaf weights; helps control model complexity.	[0,12]
gamma	Min. loss reduction required to make a split.	[0,6]

terms of performance metrics to see how well different ML and ensemble methods worked. The confusion matrix of the suggested method (i) (Proposed PSO_XGBoost) and other approaches numbered (a) Logistic Regression (LR), (b) Support Vector Machine (SVM), (c) K-Nearest Neighbor (K-NN), (d) AdaBoost, (e) XGBoost, (f) Decision Tree, (g) Random Forest, and (h) GA_XGBoost along with the test results, is shown in Figure 2 from (a) to (i). Figures 3(a)–3(i) present visual representations of the ROC curves and associated AUC values for labels 1–6. In Figure 3 (from 3a to 3i), the ROC values of every model are more excellent, which justifies that all models have the highest TP rate and lowest FP rate, especially the proposed model, which is superior to other models. It is clear from this outcome that the suggested approach in Figures 2(i) and 3(i) is practical. Figures 4 and Figure 5 show the results of the accuracy tests for all the ML and Ensemble methods considered in this study. Figure 4 shows that compared to the evaluated ML algorithms and other ensemble approaches, the proposed PSO_XGBoost model has obtained better accuracy. The suggested solution continually shows superiority in IIoT communication capabilities upon visual examination. The system's inherent robustness and efficiency effectively mitigate cyberattacks in practical IIoT settings.

A. Comparative Studies with State-of-the-art-approaches

Table IV presents a comparative analysis of our proposed PSO_XGBoost framework with existing studies using different datasets. Our model, evaluated on the X-IIoTID dataset, achieves the highest accuracy (99.80%) and F1-score (99.82%), outperforming other models. In contrast, DT applied to the MQTTset dataset achieved 99.68% accuracy, slightly lower than our model. Similarly, RF on the HRSS dataset attained 98.15% accuracy, while CPLNN trained on a synthetic dataset reached 98.50% accuracy, both trailing our approach. Notably, ANN on the SCADA dataset yielded the lowest performance, with 96.00% accuracy and 91.00% F1score, indicating its limitations in handling IoT-related attack detection. The superior performance of PSO_XGBoost can be attributed to the optimized hyperparameter selection via PSO, which enhances the model's ability to discern well across attack types. This comparison shows how well our framework works to detect intrusions in industrial IoT environments reliably and accurately.

B. Discussion on Present Research

The experimental findings demonstrate the exceptional performance of the proposed PSO_XGBoost model, attaining an impressive accuracy of 99.80%, outperforming conventional ML models (LR, SVM, KNN) and ensemble techniques (RF, AdaBoost, XGBoost, GA_XGBoost). The robustness of PSO_XGBoost is evident in its higher recall (0.9981), precision (0.9984), and ROC-AUC score (0.9994), indicating its effectiveness in determining cyber threats in IIoT environments. The confusion matrix and ROC-AUC analysis further validate the proposed model's efficiency in accurately classifying attack types with minimal misclassification errors. The findings have significant implications for enhancing IIoT security by improving intrusion detection accuracy, thereby reducing potential system vulnerabilities. The proposed approach leverages the optimization capabilities of PSO to fine-tune XGBoost parameters, leading to superior generalization and reduced overfitting compared to traditional hyperparameter tuning methods. The improved F1 and F2 scores indicate that the model balances recall and precision, making it highly decisive for real-world applications. Despite its potent performance, the research has certain limitations. Although the suggested model demonstrates computational efficiency in contrast to traditional tuning methods, the training phase could still demand significant resources when applied to extensive IIoT networks. Furthermore, the model's effectiveness was evaluated on a specific dataset (XIIoT), which could restrict its generalizability to other IIoT datasets exhibiting varying attack patterns. Future research can explore lightweight versions of PSO_XGBoost tailored for edge and fog computing environments to enhance real-time detection capabilities. Furthermore, integrating explainability techniques, such as SHAP or LIME, could provide better interpretability of the model's decision-making process. Expanding the dataset with real-time IIoT traffic and incorporating adaptive learning mechanisms could improve model adaptability to evolving threats.

6. CONCLUSION AND FUTURE SCOPE

In recent years, the increasing use of advanced technology and the simultaneous growth of the worldwide population have led to a continuous rise in expenditures for IIoT and related services. Furthermore, the expeditious progress in IoT technology has contributed significantly to the substantial growth of the IIoT, with the primary objective of improving the overall prosperity of businesses.

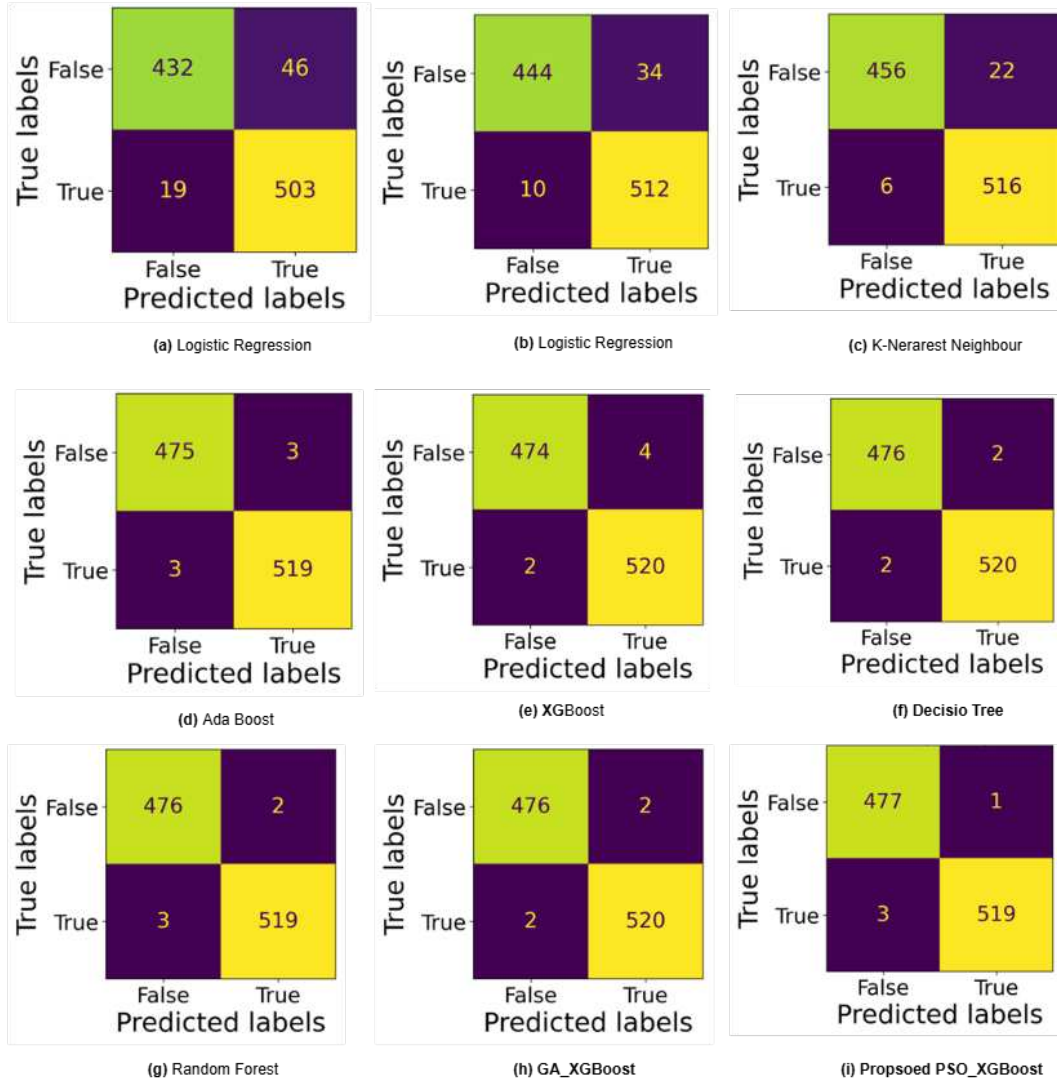


Figure 2. Confusion Matrix Analysis for all Investigated Models

TABLE III. Evolution of test-bed clusters

Classification Model	Accuracy (in %)	Precision	Recall	F1 Score	F2 Score	ROC-AUC
LR	93.67	0.9570	0.9030	0.9300	0.9140	0.9750
SVM	95.60	0.9700	0.9280	0.9520	0.9380	0.9930
KNN	97.20	0.9870	0.9530	0.9700	0.9600	0.9910
AdaBoost	99.40	0.9941	0.9942	0.9941	0.9939	0.9981
XGBoost	99.60	0.9930	0.9970	0.9950	0.9970	0.9960
DT	99.64	0.9965	0.9963	0.9964	0.9960	0.9984
RF	99.70	0.9971	0.9970	0.9970	0.9970	0.9990
GA_XGBoost	99.70	0.9974	0.9971	0.9972	0.9970	0.9986
Proposed PSO_XGBoost	99.80	0.9984	0.9981	0.9982	0.9980	0.9994

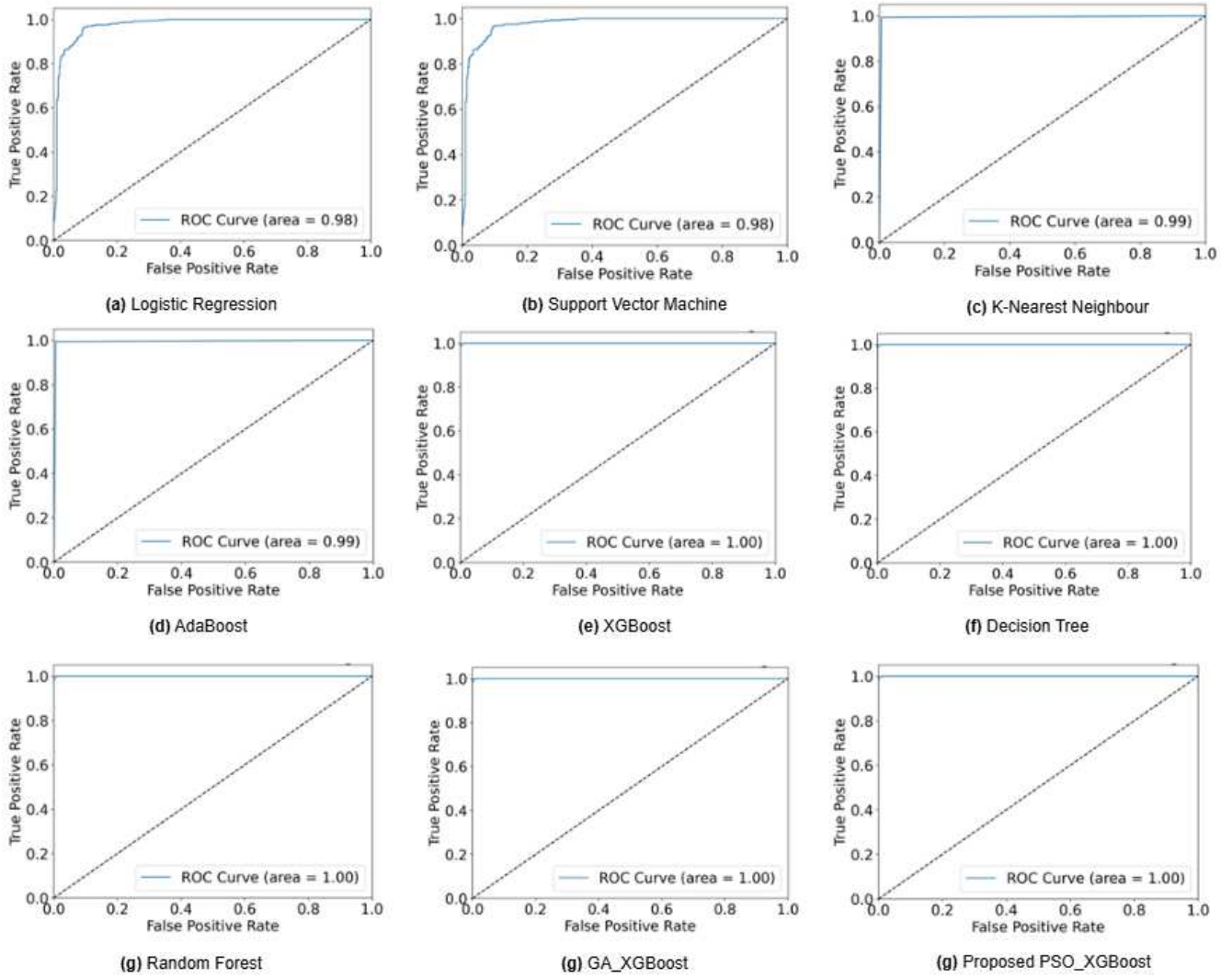


Figure 3. ROC-AUC Analysis for all Investigated Models

TABLE IV. Comparison of Proposed Framework with Existing Studies

Model Name	Dataset Name	Accuracy	F1 Score	Year, Ref.
DT	MQTTset dataset	99.68%	99.66%	2023,[11]
ANN	SCADA Dataset	96.00%	91.00%	2023,[13]
RF	HRSS dataset	98.15%	96.03%	2022,[16]
CPLNN	Synthetic Dataset	98.50%	97.49%	2021,[17]
Proposed PSO_XGBoost	XIIoTID	99.80%	99.82%	Current Work

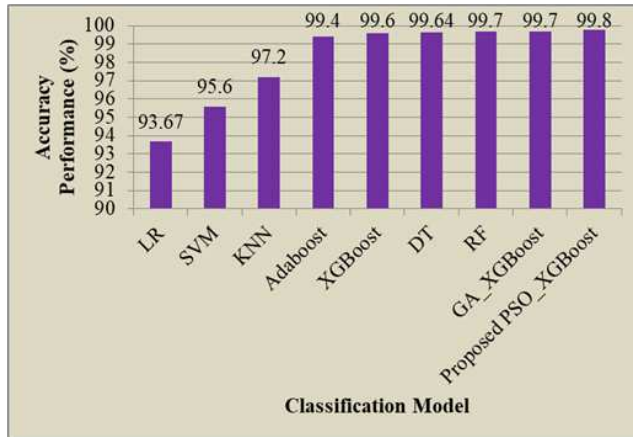


Figure 4. Accuracy Analysis all Considered Model

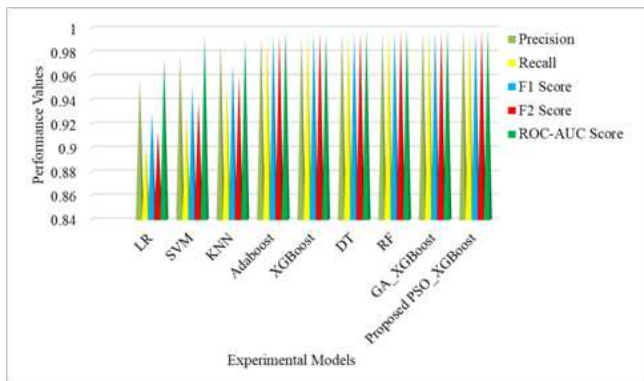


Figure 5. Comparison Analysis all Experimental Models

Hence, it is essential to include the IIoT in the sector to ensure the delivery of cost-efficient services of exceptional caliber. A significant challenge while implementing the IIoT ecosystem is the increasing cyberattack vulnerability. To strengthen the security of the IIoT ecosystem against cyber threats, several researchers are developing various techniques to strengthen its defenses against hostile cyber incursions. This study introduces an enhanced approach employing PSO_XGBoost ensemble learning to simultaneously increase the accuracy of Intrusion Detection Systems (IDS) and safeguard the security of Industrial Internet of Things (IIoT) systems. Furthermore, the findings and evaluations consistently demonstrate that the proposed model has achieved a precision rate of 99.80% compared to alternative models. ML techniques, DL methodologies, and complex algorithms could enhance the model.

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